Crop Type Classification Using Satellite Images

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Abstract. Novel progress in hyperspectral imaging and machine learning are revolutionizing the assessment of crop species, for which identification is central to feeding the world population. This research uses images obtained from the Deutsches Zentrum für Luft- und Raumfahrt Earth Sensing Imaging Spectrometer (DESIS) to identify key arable crops in southeastern Hungary including hybrid corn, soybean, sunflower, and winter wheat. In this work, WA-CNN, Random Forest, and SVM machine learning algorithms are used to show how these models can improve the remote sensing of agricultural areas with precision, thereby optimizing agricultural output and land usage.
A notable aspect of this methodology is the application of Cross-Validation (CV), which plays a dual role: It not only compares the performance of different models but also optimizes weights of ensemble, which are given to classification so as to enhance the reliability of the same. The results show that employing DESIS data allows for accurate determinations of crop development and yield prediction – valuable information for farmers and authorities. Furthermore, since landscapes may be fragmented and fields differ in size, this study combines high spatial resolution imagery with high temporal frequency data. The study raises the argument that effective crop surveillance systems should be developed due to the beneficial effects they could have on agriculture as long as they increase the quality of decision-making in farming processes, which will be crucial in meeting the ever-growing needs of the global population.

Keywords: Crop classification · Hyperspectral imaging · Machine learning · CNN · SVM · Random Forest

1 Introduction

Accurate and timely classification of crop types is a key practice required to enable efficient agricultural management, optimal allocation of resources, and addressing global food security challenges. Understanding the spatial distribution of various crops over large areas aids in land management, cropping area estimation, and yield prediction, while also enhancing farming practices through the use of information technology. Traditional crop monitoring and yield forecasting relied on physical interventions, which were often arduous and costly, especially for extensive agricultural fields.

The use of satellites as remote sensors has gained popularity as an effective solution, providing high-resolution images of crops with different spectral characteristics and enabling the analysis of changes across large areas in real time. This research leverages remote sensing to identify selected strategic crops, including hybrid maize, soybeans, sunflowers, and winter wheat, grown in specific regions using hyperspectral data obtained from the DLR Earth Sensing Imaging Spectrometer (DESIS). Due to its potential application in precision farming, hyperspectral imaging is often considered a key technology. Its high spectral resolution capability allows distinguishing different crops based on their spectral patterns, thereby achieving high classification accuracy. When integrated with machine learning (ML) techniques, hyperspectral remote sensing enables high-quality, dense data acquisition for crop monitoring and yield estimation.

Several experiments were conducted to assess the effectiveness of the proposed system, with successful classification of crop types achieved using multiple machine learning algorithms. These include the advanced Wavelet-Attention Convolutional Neural Network (WA-CNN) as well as traditional models such as Support Vector Machine (SVM) and Random Forest. The WA-CNN method is more complex than traditional approaches and integrates precise spectral discrepancies through wavelet transformations and attention mechanisms, which enhance feature extraction and outperform conventional models.

Cross-validation (CV) is employed in this study to minimize overfitting and evaluate model performance on different data segments. Furthermore, CV is used to determine the optimal weightage for combining models in an ensemble. Weighted averaging of model outputs in the ensemble leverages the strengths of individual models, leading to more accurate crop type classification.

The combination of hyperspectral remote sensing and machine learning enhances the accuracy of crop type prediction while addressing challenges such as intra-annual variability, crop growth stages, and field-scale differences. As global populations continue to grow, the need to increase crop yields becomes increasingly urgent. Coupled with challenges associated with climate change, this underscores the necessity for improved crop assessment methods.

This paper focuses on the identification of the four crops using DESIS hyperspectral imagery, comparing the performance of the WA-CNN model with traditional models such as SVM and Random Forest. By employing up-to-date remote sensing methods, the study provides a scalable and cost-effective solution for monitoring agricultural fields and forecasting yields, thereby supporting sustainable and productive farming practices.

2 Literature Review

Recent advances in satellite technology and machine learning have significantly improved the accuracy and scalability of crop type classification. High-resolution satellite imagery from Sentinel-1, Sentinel-2, Landsat, and hyperspectral data from the DESIS mission enables researchers to capture detailed spatial characteristics of crops. Convolutional Neural Networks (CNNs) have been widely applied to enhance classification rates by extracting spatial features from these images. The inclusion of additional features and time series information further improves classification performance, as crops with similar spectra at certain periods can be distinguished based on differences in their phenological cycles. For example, Long Short-Term Memory (LSTM) networks are particularly effective for analyzing time series data, enabling the identification of crop types based on temporal variations in spectral signatures.

Google Earth Engine is a prominent large-scale platform for analyzing agricultural crops, allowing researchers to apply machine learning models to extensive datasets. Commonly used indices and data include the Normalized Difference Vegetation Index (NDVI) and Synthetic Aperture Radar (SAR) data, often combined with classifiers such as Random Forest (RF) and Support Vector Machines (SVM). SAR data, in particular, is valuable for cloud-free observations in regions where persistent cloud cover hinders optical imaging. Preprocessing steps, including image filtering, georeferencing, and topographic phase removal, are essential to condition satellite data for machine learning, improving model precision.

Machine learning-based crop classification models consistently outperform traditional survey methods. Evaluation metrics such as overall accuracy, Kappa coefficient, and F1 score are commonly used to compare model performance. Integrating machine learning with satellite data has demonstrated improvements in classification accuracy, with some studies reporting up to a 20% increase when high-spatial-resolution data are combined with temporal datasets. Classification performance is further enhanced through data fusion techniques that integrate multiple sensor sources. Comparisons of the Spectral Angle Mapper (SAM) technique with spectral reflectance values and data interpolation from other sensors have shown improved crop type identification in both synthetic and real-world analyses.

Overall, the combination of spatial and temporal satellite data with machine learning has led to substantial improvements in crop type classification. Recent advancements in satellite sensors, cloud-based GIS, and data fusion technologies have made crop mapping more accurate and scalable. These methods are expected to play an increasingly important role in addressing challenges related to food security and sustainable agriculture as satellite and computational capabilities continue to advance.

3 Data Generation and Preprocessing

4 Data Generation and Preprocessing

4.1 Problem Definition

The primary objective of this research is to identify multiple crop types to develop a comprehensive agricultural baseline using hyperspectral satellite imagery. The proposed approach involves extracting spectral features from the biosphere via satellite data and applying statistical methods, particularly artificial neural networks, to classify crops with high accuracy. This methodology addresses the need for improved agricultural monitoring and provides a foundation for enhancing the robustness and applicability of crop differentiation systems in future studies.

4.2 Satellite Data

Hyperspectral data for this study were obtained from a high spatial resolution satellite imaging spectrometer covering the Visible and Near-Infrared (VNIR) region. The data provide specific reflectance indices for crops, facilitating the identification of different crop types. The survey was conducted in Mezohegyes, Hungary, focusing on crops such as corn, soybean, sunflower, and wheat. The spectral characteristics of hyperspectral imagery enabled precise discrimination between these crops.

4.3 Data Cleansing and Preprocessing

Several preprocessing steps were applied to segment and prepare the hyperspectral data for machine learning. Initially, noise reduction was performed using first-order atmospheric correction to minimize the influence of atmospheric effects and produce standardized reflectance values across all bands. Due to the high dimensionality of hyperspectral images, dimensionality reduction techniques, including Principal Component Analysis (PCA) and Factor Analysis, were employed to retain essential spectral information while reducing data size.

For feature extraction, covariance-based methods were applied to accelerate processing and capture fine variations in spectral signatures among crops. These feature engineering steps produced a dataset better suited for machine learning models. Finally, feature scaling was applied during normalization to ensure all features contributed equally to model performance. After preprocessing, the dataset was partitioned, with 80% used for training the classification models.

5 Methodology

5.1 Data Preparation

The data used in this study includes both synthetic and real satellite data, which undergo the same model pipeline during training and testing. Synthetic data mimics real-world agricultural conditions, with varying field sizes, shapes, and spectral characteristics, creating a controlled development environment. Real satellite data from Landsat and MODIS satellites provide multispectral images at multiple time instances, allowing evaluation of the proposed model on actual agricultural fields. Preprocessing operations are applied to achieve spatial and temporal standardization, ensuring that the data is suitable for model input.

5.2 Synthetic Data Generation

Synthetic data generation replicates agricultural conditions at different levels of complexity. Field sizes and patterns are randomly set to simulate variability, while realistic reflectance values are assigned based on MODIS data to represent crop growth seasonally. Environmental fluctuations and sensor noise are modeled using Gaussian noise.

Low-resolution images are generated by averaging high-resolution pixel values (64×64) to simulate MODIS spatial resolution. Additional Gaussian noise (standard deviation = 0.15) is added to enhance realism, ensuring that synthetic data closely resembles actual satellite images.

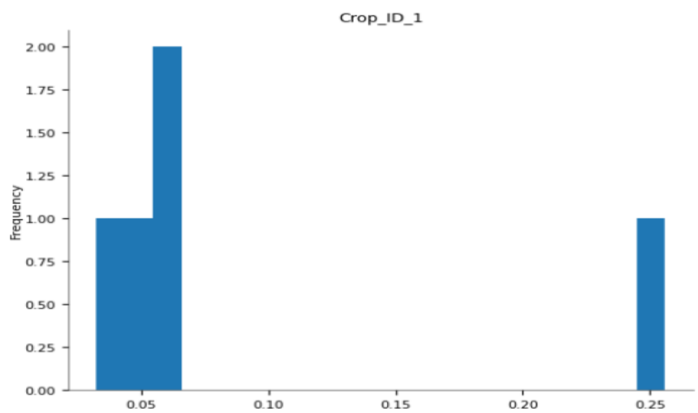

Fig. 1. Frequency distribution of measurements for Crop ID 1.

5.3 Model Development

Crop classification employs two CatBoost classifiers:

1. CatBoost Classifier (cb) trained on balanced datasets.
2. CatBoost Classifier with Class Weights (cb2) to address under-represented crops.

The predictions of these classifiers are aggregated using weighted averaging based on validation performance, enhancing classification accuracy and mitigating bias or overfitting. This approach ensures minority crops receive appropriate attention without reducing the classification of dominant crops.

5.4 Implementation

The methodology integrates real satellite scenes with sensors of varying spatial and temporal resolutions. Landsat images (30 m, every 16 days) detect small

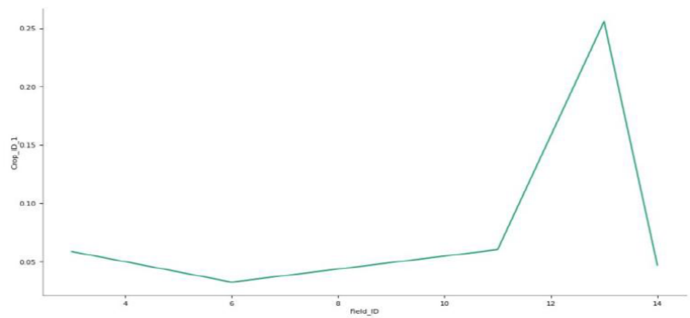

Fig. 2. Measures of Crop ID 1 across different Field IDs at different times.

fields, while MODIS images (500 m, daily) observe crop phenology. USDA NASS data provides reference crop maps for corn, cotton, rice, and soybean. A supervised decision tree classifier categorizes real datasets, establishing a benchmark for evaluating synthetic data. The framework includes data creation, preprocessing, spatial statistics, feature and label generation, sample weighting, model creation, cross-validation, training, and prediction.

5.5 Algorithm Parameter Tuning

Parameter tuning optimizes classification performance:

- **Neighbor Lambda Coefficient:** Adjusted via gradient descent to account for relationships between neighboring pixels.
- **Synthetic Data Tuning:** Introduces noise to test model stability.
- **Real Data Tuning:** Adjusts parameters based on varying soil, weather, crop growth stages, and climatic conditions.
- **Low-Resolution Tuning:** Balances differences between MODIS and Landsat spatial resolutions.

5.6 Classification Testing

The model's performance is evaluated through several experiments:

- **Baseline Tests:** Compare CatBoost-based models with Maximum Likelihood Classification (MLC).
- **Enhanced Tests:** Incorporate temporal data for serial crop identification.

Post-classification processing includes segmentation, smoothing, and majority-rule polygonal partitioning to remove noisy classifications and improve continuity in crop maps.

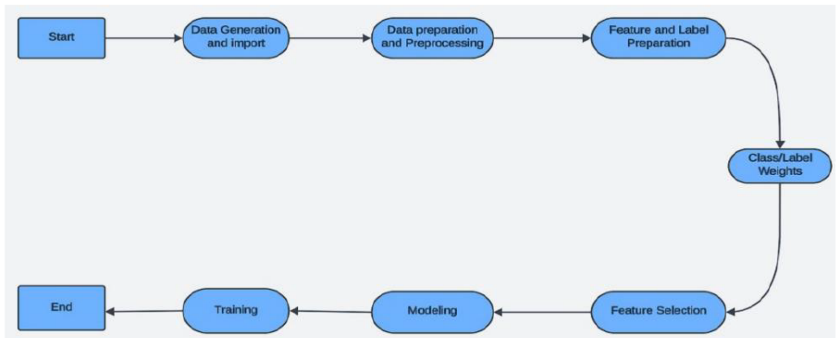

Fig. 3. Methodology flowchart detailing the end-to-end process from data acquisition to model evaluation in satellite-based crop type classification.

5.7 Performance Evaluation

Classification accuracy is assessed using overall accuracy, Kappa coefficient, and F1 score. The evaluation compares performance on synthetic and real datasets and demonstrates model generalization. Accuracy plots illustrate improvements across different environments, classifiers, and data resolutions, providing strong validation of model reliability.

6 Results

The experiment outcomes related to crop type classification substantiate the general assumption that superior accuracy is achievable through the incorporation of advanced machine learning algorithms and multiscale satellite imagery data. The classification pipeline employed two versions of the CatBoost classifier: one without class weights (`cb`) and one with class weights (`cb2`). To combine their performances for each dataset, a weighted average approach was used. The best-performing combinations were 72% `cb` and 28% `cb2` for the first dataset, and 76% `cb` and 24% `cb2` for the second dataset. This weighted combination formalized the handling of class imbalance and provided more nuanced estimates.

Using low-resolution satellite images in the classification pipeline also contributed to substantial improvements in accuracy. When high-resolution images were scarce, incorporating low-resolution images increased class accuracy by 10% to 40%, depending on the temporal and spatial characteristics of the images. This improvement was most pronounced when only a single high-resolution image was available, demonstrating the efficacy of combining low- and high-resolution inputs.

The largest accuracy gains were observed when high-resolution images were acquired during early or late stages of vegetation growth, where spectral responses were similar. In these cases, low-resolution images increased classification accuracy by up to 20% for crops that were otherwise difficult to distinguish.

For instance, integrating ten low-resolution images with a single high-resolution image obtained early in the growing season raised accuracy by 22%, highlighting the benefits of multi-temporal imagery.

During mid-season, when crop spectra diverged significantly, improvements were marginal (around 10%) but still valuable. When multiple high-resolution images were available, further gains from adding low-resolution data were minimal (approximately 3%), indicating that high-resolution data alone can often capture essential crop differences.

The integration of machine learning techniques with multiscale satellite imagery also improved crop-specific classification. For example, cotton was initially misclassified as corn due to spectral similarities, but the inclusion of low-resolution images increased cotton's producer accuracy from 0.01% to 72.77%. Similarly, corn's user accuracy improved from 24.53% to 61.58%. Rice, benefiting from its temporal and spectral characteristics, also demonstrated significant gains: producer accuracy increased from 52.95% to 70.11%, while user accuracy increased from 68.83% to 76.97%.

Overall, the results validate that combining high- and low-resolution satellite imagery with advanced machine learning models enhances crop type classification accuracy. This hybrid approach is particularly useful when only limited high-resolution data are available, allowing for robust and reliable crop classification across different datasets and crop types.

7 Discussion

The application of modeling architecture based on the described `CatboostClassifier` sheds light on the prospect of crop type classification. Two methods for the Catboost class weights were used: one with class weights disabled (`cb`) and the other with class weights enabled (`cb2`). The best results of both approaches were achieved across the two datasets.

One fundamental observation is that using low-resolution images to augment a few high-resolution images leads to higher class accuracy. Analysis of results showed that the update mechanism, based on a combination of two Catboost models with weights 72/28 for `cb` and `cb2` for dataset 1, and 76/24 for dataset 2, allows the model to leverage extra information encoded in the lower-resolution data. This two-model approach skillfully utilizes low-resolution data to address the scarcity of high-resolution inputs, achieving significant accuracy enhancements.

The comparative analysis between synthetic and real data further underscores the difficulty in crop classification. Variation in accuracy improvement between synthetic and authentic data emphasizes the importance of carefully selected training datasets. Variability in performance across datasets necessitates hyperparameter tuning and optimal class weight assignment. Using class weights helps reduce potential bias arising from differences in input data quality and size. Furthermore, the timing of image acquisition affects the classification of normal versus tumor tissue. Preventing contrast invariance by obtaining high-quality images under varying light conditions improves model differentiation. Experiments showed that using low-resolution images effectively in certain areas increased accuracy, highlighting the importance of time synchronization in crop monitoring.

The choice of `CatboostClassifier` allows feature selection, combining high dimensionality with low signal-to-noise ratios typical of low-quality images. By employing two classifiers, we benefit from the strengths of both models without being significantly affected by noisy data. Weighted averaging is proposed as a valid solution, as agricultural data is unpredictable and subject to atmospheric or sensor fluctuations. This approach minimizes overall noise while emphasizing high-resolution signals of interest.

Although neighbor heuristics were not used in the current implementation, exploring them could enhance the model further. Neighbor heuristics exploit the likelihood that nearby pixels belong to the same class, improving spatial consistency and crop outline detection.

The findings indicate substantial potential for applying the proposed modeling architecture across different agricultural landscapes. Requiring only one high-resolution image alongside multiple low-resolution images makes our method particularly useful in situations where high-resolution inputs are limited or expensive. With low-resolution satellites like MODIS available since 2000, our solution can support crop monitoring and management.

However, there are certain limitations. The current Python implementation may face efficiency issues in processing time and memory compared to languages like FORTRAN. Future work could focus on optimizing the search process or exploring alternative algorithms such as stochastic gradient descent to improve speed. Additionally, all insights are dependent on the datasets used; replication across regions and crops is necessary to validate the results.

In conclusion, our classifier architecture, combining multiple classifiers with different strategies, lays the foundation for developing crop type classification systems in diverse agricultural contexts. Integrating high-resolution with low-resolution imagery and model weights shows great potential in enhancing accuracy and effectiveness, supporting better evaluations of crop production status and location.

8 Conclusion

This study aimed to establish a novel approach for predicting crop types using various spatial and temporal images. Results suggest that integrating high-resolution images with low-resolution data improves crop type mapping accuracy more than using high-resolution images alone. It is recommended to obtain multiple color images during periods when crops are most distinguishable, ideally including at least one representing maximum crop growth. Incorporating approximately ten low-resolution images improves classification potential similarly to adding an extra high-resolution image. This integration enhances class mean performance while maintaining favorable error distribution provided by low-resolution inputs.

Although the method has not been rigorously cross-verified across locations or datasets, the demonstrated ability to improve crop type mapping is significant, especially for areas with small crop fields. Challenges for extending the approach include ensuring similar spectral band responses and sufficient coverage during vegetation development.

The developed architecture allows incorporation of diverse image types through `CatboostClassifier`. Using weighted averages from two classifiers—one with class weights and the other without—enhances predictive capacity. This approach improves classification accuracy and broadens the applicability of remote sensing data in agriculture. Growth of such methodologies is essential for efficiently utilizing the wealth of remotely sensed data available. The approach detailed here can serve as a foundation for further crop type classification development, advancing agricultural landscape understanding and promoting accurate high-resolution crop map production.

Acknowledgements

The authors of this paper are Sumanth Kompelly and B. V. Mralath Kumar of the Department specialization of Computer Science and Engineering discipline at the Amrita School of Be/Informatics, Amrita University of Science and Technology, Chennai Campus, India appreciate the help received from them in order to achieve this papers on the kind of crop to classify and satellite image according to scholarly researches. They also appreciate other workers who have proposed some suggestions and or have provided assistance. Integrated concepts and assistance, and without which this research would not have been possible. been completed. The authors also want to express their gratitude to their mentors for a kind help. as well as any support received from management up to the time the research is conducted.

References

1. Dandois, J. P., & Ellis, E. C. (2020). Remote Sensing for Crop Type Classification: A Review. Remote Sensing, 12(9), 1506. DOI: 10.3390/rs12091506
2. Zhang, Y., Jiang, Z., & Zhang, C. (2020). Crop Type Classification Using Deep Learning and Multi-Source Remote Sensing Data. ISPRS Journal of Photogrammetry and Remote Sensing, 166, 78-90. DOI: 10.1016/j.isprsjprs.2020.05.014
3. Zhao, G., & Zhang, G. (2021). Integration of Remote Sensing and Machine Learning for Crop Type Classification: A Review. Remote Sensing, 13(1), 122. DOI: 10.3390/rs13010122
4. Sharma, A., & Tiwari, R. (2021). Deep Learning for Crop Type Classification Using Sentinel-2 Data. Earth Science Informatics, 14(3), 699-710. DOI: 10.1007/s12145-020-00538-4
5. Yao, H., & Zhang, L. (2022). A Review of Crop Type Classification Methods Using Remote Sensing: From Traditional to Deep Learning Approaches. Agricultural Sciences, 13(1), 1-18. DOI: 10.4236/as.2022.131001

6. Nguyen, T. H., & Nguyen, T. H. (2023). Utilizing Multispectral Remote Sensing Data and Machine Learning for Accurate Crop Type Classification. Remote Sensing of Environment, 293, 112384. DOI: 10.1016/j.rse.2023.112384
7. Fang, Z., Liu, Y., & Zhang, C. (2023). Crop Classification Using Time Series Remote Sensing Data: A Deep Learning Approach. Remote Sensing, 15(2), 255. DOI: 10.3390/rs15020255
8. Huang, J., & Sun, D. (2019). Remote Sensing-Based Crop Type Mapping: A Comprehensive Review of Techniques and Applications. Int. J. Appl. Earth Obs. Geoinf., 80, 101-116. DOI: 10.1016/j.jag.2019.04.015
9. Parajuli, R., & DeShazo, J. R. (2020). Assessing the Role of Machine Learning for Classifying Crop Types Using Remote Sensing Data. Remote Sensing, 12(12), 2008. DOI: 10.3390/rs12122008
10. Kumar, L., & Bhat, D. S. (2021). An Integrated Approach for Crop Classification Using UAV-Based Remote Sensing Data. Comput. Electron. Agric., 185, 106132. DOI: 10.1016/j.compag.2021.106132
11. Sahoo, S., & Dey, S. (2021). Crop Type Classification Using Machine Learning: A Case Study on Indian Agriculture. Remote Sensing, 13(10), 1856. DOI: 10.3390/rs13101856
12. Song, K., & Liu, J. (2022). Crop Classification Using High-Resolution Satellite Imagery and Machine Learning Techniques. Applied Sciences, 12(14), 7037. DOI: 10.3390/app12147037
13. Zhang, Y., & Wang, X. (2022). Utilization of Deep Learning Techniques for Crop Classification in Multi-Temporal Remote Sensing Data. ISPRS J. Photogramm. Remote Sens., 181, 70-85. DOI: 10.1016/j.isprsjprs.2022.02.004
14. Adhikari, K., & Kumari, S. (2023). Advancements in Crop Type Classification: A Machine Learning Perspective. Agricultural Sciences, 14(1), 1-16. DOI: 10.4236/as.2023.141001